ATLAS-R

ROCKET & CANSAT INTEGRATED SYSTEM

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INTRODUCTION

This project focuses on two main aspects: rocket design and materials, which contribute to the rocket's structural integrity, durability, and flight stability, and avionics and recovery systems, which enable flight monitoring, sensor data collection, and the safe recovery of the rocket and its CanSat payload.

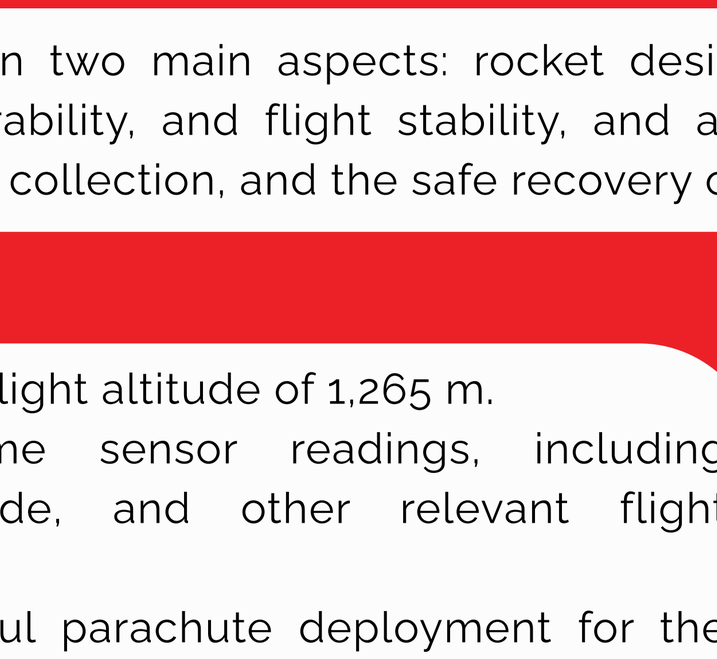
OBJECTIVES

- To achieve a target flight altitude of 1,265 m.
- To obtain real-time sensor readings, including temperature, altitude, and other relevant flight parameters.
- To ensure successful parachute deployment for the safe recovery of the rocket and its CanSat payload.

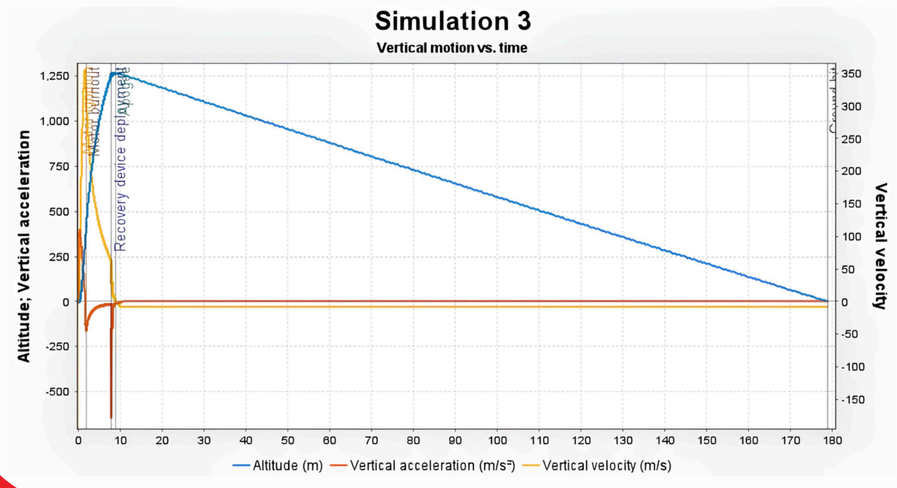
KEY INNOVATIONS

1. Compact Binary Telemetry
2. Bidirectional Communication with Acknowledgment
3. Dual Mode Recovery Control
4. Integrated Multi Sensor Flight Data Acquisition

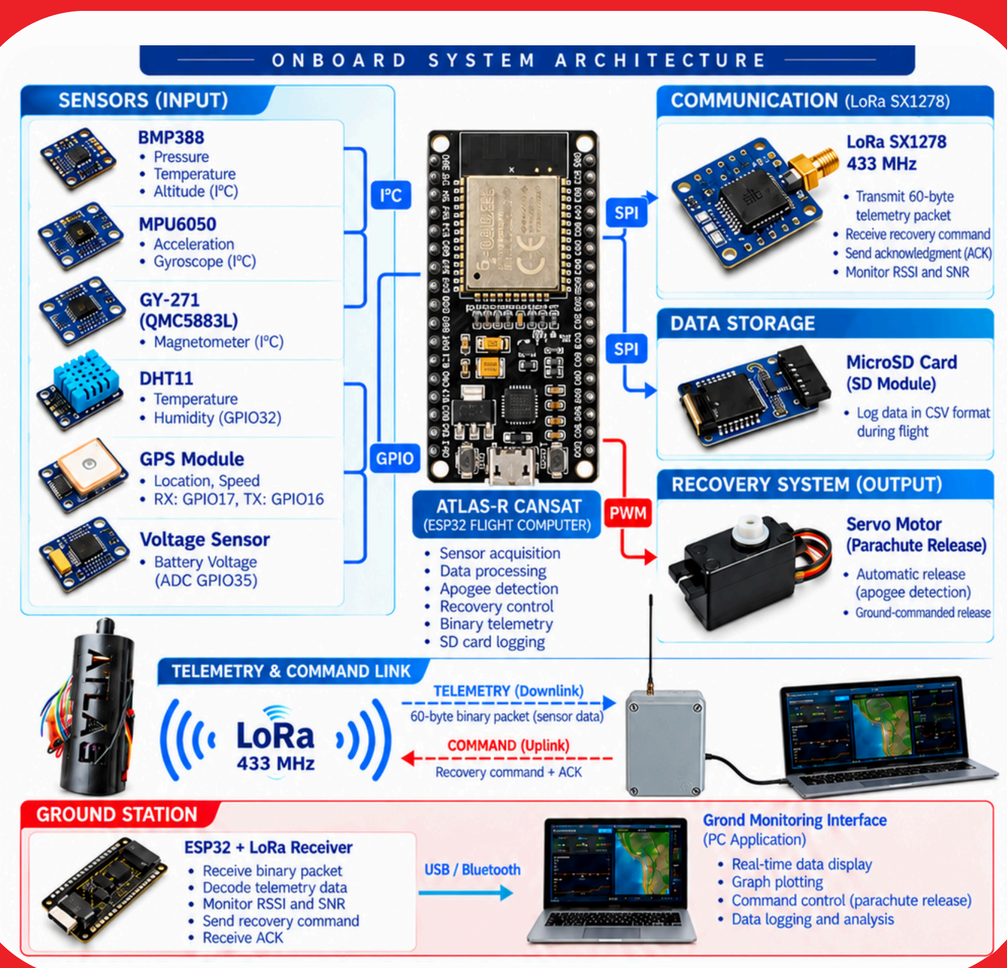
DESIGN



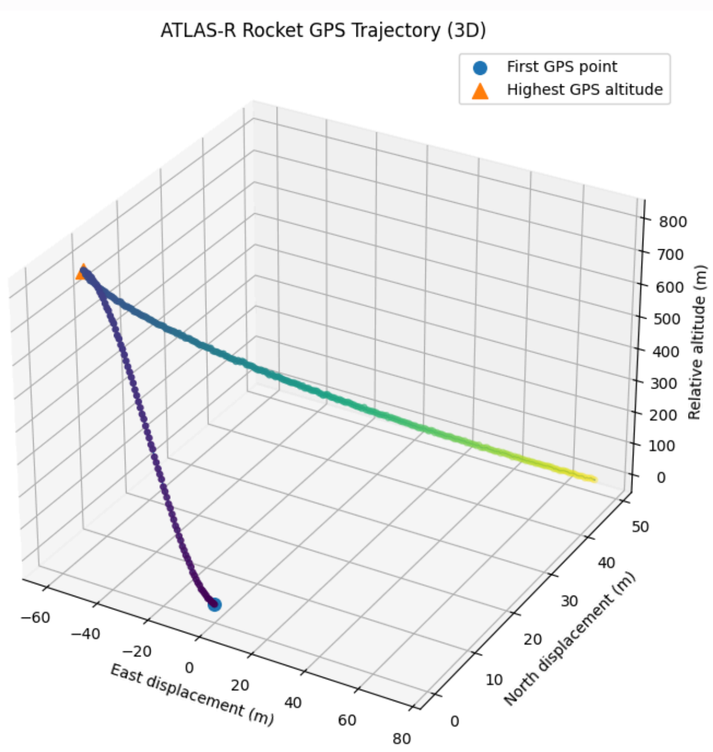
- Expected apogee: 1,250 – 1,265 m
- Maximum velocity: ~ 355 m/s
- Peak acceleration: ~ 396 m/s²
- Stability margin: 1.32 calibers

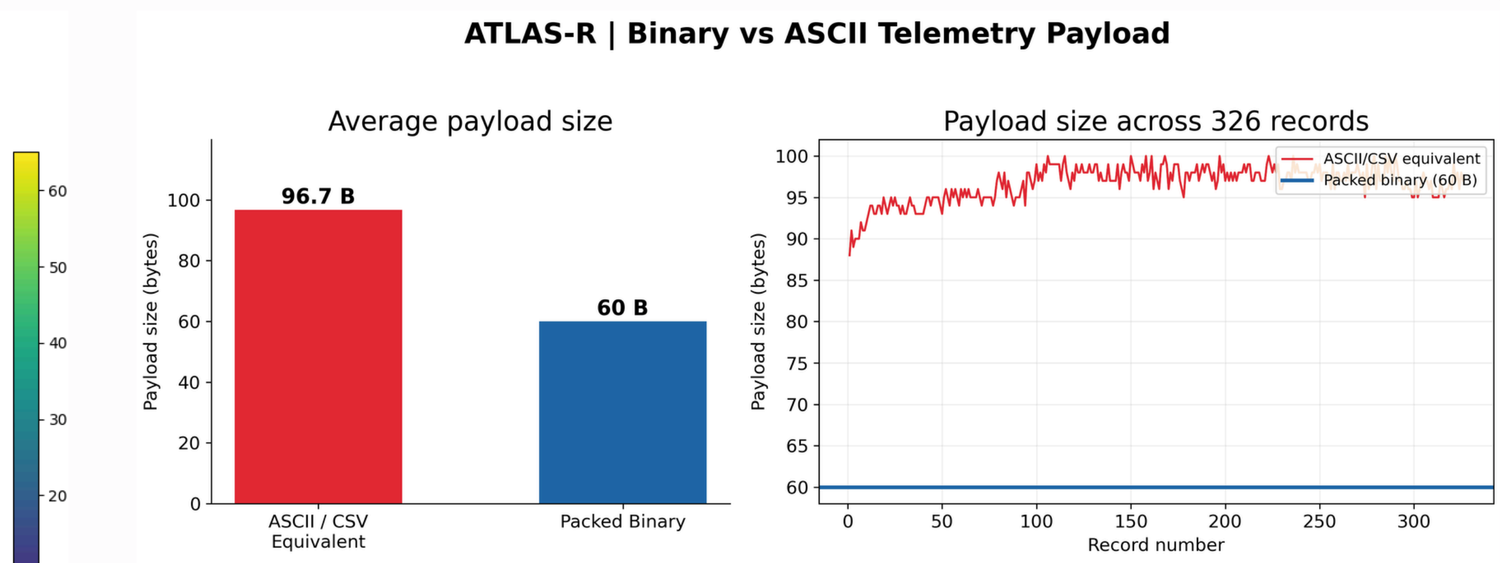


AVIONICS ARCHI & TELEMETRY



RESULTS





- Max altitude: 811.25 m
- Maximum velocity: ~ 54.07 m/s
- Peak acceleration: ~ 34.32 m/s²
- Acceleration magnitude: 3.5g

